

Question Number	Answer	Additional Guidance	Mark
3(a)(i)	all cylinders same {length / diameter / size} / temperature kept constant (1)	Accept same volume and surface area	(1)

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	repeat measurements {and obtain mean / SD / identify anomalies} for each concentration / use callipers to measure length changes / description of use of ruler to avoid parallax errors (1)	Candidates are going to express their answers in many different ways	(1)

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	An answer including the following points: - the liquids should be mixed in ratio 2:3 (1) - The 3 is water (1)		(2)

Question Number	Answer	Additional Guidance	Mark
3(b)(ii)	An answer including the following points: - it {dissolves (completely in water) / is soluble} (1) - it is {not broken down / not metabolically active / inert} (1) - it {is not taken up by the carrot / does not cross membrane of cells / has osmotic effect} (1)		(2)

Question Number	Answer	Additional Guidance		Mark														
3(c)(i)	A table drawn with the following features: - suitable table drawn (1) - headings with units (1) - all data correctly entered (1)	<table><tr><td>concentration (of sucrose solution) / mol dm⁻³</td><td>change in length at 50 min / mm</td></tr><tr><td>0.0</td><td>4.0</td></tr><tr><td>0.2</td><td>1.6</td></tr><tr><td>0.4</td><td>-1.4</td></tr><tr><td>0.6</td><td>-3.4</td></tr><tr><td>0.8</td><td>-4.6</td></tr><tr><td>1.0</td><td>-5.2</td></tr></table>		concentration (of sucrose solution) / mol dm ⁻³	change in length at 50 min / mm	0.0	4.0	0.2	1.6	0.4	-1.4	0.6	-3.4	0.8	-4.6	1.0	-5.2	(3)
		concentration (of sucrose solution) / mol dm ⁻³	change in length at 50 min / mm															
		0.0	4.0															
		0.2	1.6															
		0.4	-1.4															
		0.6	-3.4															
		0.8	-4.6															
		1.0	-5.2															

Question Number	Answer	Additional Guidance	Mark
3(c)(ii)	An answer including the following steps : • reading values from a tangent (1) • correct answer (in mm min⁻¹) (1)	e.g. -3.6 at 10.0 to -4.8 at 20.0, diff is 1.2 1.2 ÷ 10 = -0.12 (mm min⁻¹)	(2)

Question Number	Answer	Additional Guidance	Mark
3(c)(iii)	An answer including the following points : - the cylinder stops decreasing in length (1) - because all the water (that can be lost) has been lost / water potential inside = water potential outside (1)	Candidates will express mark point 2 in a number of ways	(2)

Question Number	Answer	Additional Guidance	Mark
3(c)(iv)	An answer including the following points - the cylinder stops increasing in length (1) - as the cells are fully turgid / water potential inside = water potential outside (1)	Candidates will express mark point 2 in a number of ways	(2)